

stm32_devboard_routed

Design Report

kicad-tools 0.13.0

Rev 1 | 2026-06-08 | jlcpb

Board Summary

Property	Value
Layers	2 copper (F.Cu, B.Cu)
Footprints	17 (0 SMD, 0 THT, 17 other)
Nets	12
Traces	390 segments
Vias	25
Board Size	60.0 x 40.0 mm

Design Overview

Theory of Operation

STM32F103C8 Development Board

End-to-end design example

Demonstrates circuit blocks API

Power Architecture

Power Rails: +3V3, +5V, GND, PWR_FLAG

Regulator	Device
U1	AMS1117-3.3

Assembly Notes

1 fine-pitch component; 1 polarized component

- **Fine-pitch components:** 1 (U2)
- **Polarized components:** 1 – check orientation markings

ERC Status

Metric	Count
Errors	0
Warnings	0

Status: SKIPPED – ERC skipped by user request

Bill of Materials

Value	Package	Qty	References
100nF	C_0805_2012Metric	5	C3, C12, C13, C14, C15
10uF	C_0805_2012Metric	2	C1, C2
20pF	C_0805_2012Metric	2	C10, C11
4.7uF	C_0805_2012Metric	1	C16
USER	LED_0805_2012Metric	1	D1
SWD-6	PinHeader_1x06_P2.54mm_Vertical	1	J1
10k	R_0805_2012Metric	1	R2
330R	R_0805_2012Metric	1	R1
AMS1117-3.3	SOT-223-3_TabPin2	1	U1
STM32F103C8T6	LQFP-48_7x7mm_P0.5mm	1	U2
8MHz	Crystal_HC49-4H_Vertical	1	Y1

DRC Status

Metric	Count
Errors	3
Warnings	48
Blocking	3

Status: FAIL ### Violations by Type

Violation Type	Count
pad_grid	48
via_in_pad	3
connectivity	2

Manufacturing Readiness

Verdict: NOT_READY

Action Items

- [CRITICAL] Fix 3 blocking DRC violations (via_in_pad (3))
- [CRITICAL] Increase min via drill: 0.150mm < 0.300mm required
- [OPTIONAL] Verify zone fill in KiCad: 1 nets appear incomplete but may be connected via zone fills
- [OPTIONAL] Verify zone fill in KiCad for 3 zone-connected nets
- [OPTIONAL] Review 48 DRC warnings

Routing Status

Metric	Value
Signal Net Completion	88.9% (8/9)
Overall Completion	83.3%
Complete Nets	10 / 12
Zone-Connected Nets	3
Incomplete Nets	2
Unconnected Pads	3

Zone-Connected Nets

- +3.3V
- +5V
- GND

Unrouted Signal Nets

- NRST

Unrouted Signal Nets

- NRST

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~0.88 USD
Components (estimated)	~1.36 USD
Assembly (estimated)	~2.02 USD
Total (estimated)	~4.26 USD
Batch Quantity	5
Batch Total (estimated)	~21.3 USD